BY ORDER OF THE
SECRETARY OF THE AIR FORCE

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Flying Operations

C-130 MASS OPERATIONS PILOTS CHECKLIST

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This checklist establishes procedures for employing the Modular Aerial Spray System (MASS) on C-130 aircraft employed by Mobility Air Forces (MAF) to accomplish their worldwide mission. This checklist complements AFMAN 11-2C-130V3, ADDENDA C *Modular Aerial Spray System (MASS) Procedures*, and is printed on standard 8 ½" x 11" bond paper, and trimmed to fit the standard plastic aircrew checklist binders. This checklist is intended to provide MASS certified crewmembers quick reference to procedures required for the safe execution of MASS ground and flight operations. All MASS certified C-130 pilots will carry this annex.
Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF IMT 847, Recommendation for Change of Publication; route AF IMTs 847 from the field through the appropriate chain of command.

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PART ONE – BRIEFING GUIDES

AERIAL SPRAY AIRCRAFT COMMANDER BRIEFING

1. Mission objectives
   a. Check rides
   b. Upgrade/training
2. Sequence of events
   a. Pest
   b. Pesticide
      i. Total chemical loaded
      ii. Application rate
      iii. Flow rate
   c. Route to area
   d. Spray area
      i. Charts
      ii. Acreage
      iii. Sensitive areas
      iv. No spray areas
      v. Spray on time
   e. DGPS settings
      i. CDI
      ii. AOI
      iii. Swath width
   f. Swath length
   g. Swath heading
   h. Altitude
   i. Groundspeed
   j. Swath pattern
   k. Terrain
   l. Hazards to flight
   m. Flush and/or purge
3. Weather
   a. Winds
      i. Label limit
      ii. Offset
   b. Cloud cover
   c. Visibility
   d. Humidity
   e. Inversion
4. Call sign
5. Times
   a. Show
   b. Stations
   c. Takeoff
   d. Enroute
   e. Spray start
   f. Total spray on time

g. Mission end time
6. Fuel requirements
7. MASS
   a. Mass configuration (SP1, SP2, SP3)
   b. Boom
   c. # Nozzles
   d. Orientation
8. Intelligence
   a. Threat
   b. Location
   c. Avoidance criteria
9. Interphone and radio
   a. Critical phases of flight
   b. Communication plan
10. Emergency procedures
   a. Ground egress
   b. Exits
   c. Assembly location (upwind)
   d. Chocks
   e. MASS emergency
   f. Pesticide toxicity
   g. Aircrew responsibilities
      i. Kill switch
      ii. Leak/spill (loadmaster)
   h. Emergency dumping
   i. Pax
   j. LPU's
11. Passenger/cargo load
   a. Form F
   b. Weight
12. Maintenance status
13. Personal equipment
   a. PPE
   b. Helmets
14. Crew coordination
   a. Time out
   b. Combat entry/exit
   c. Flight callouts
      i. Altitude deviation
      ii. Airspeed deviation
      iii. Swath deviation
      iv. On/off swath
      v. Towers/obstructions
15. Notes
   a. Special Interest Items
   b. Low pitch stop check for short field
   c. Slow brake release
   d. Bank angles on and off swath
   e. Chemical transfer and flaps
      Extended downwind and/or nose down

MISSION COMMANDER BRIEFING GUIDE

1. Pre-mission
   a. Base
   b. Target pest
   c. Spray areas
      i. Number of blocks
      ii. Acreage
   d. Pesticide and diluents
      i. Amount
      ii. Supplier
      iii. Location and delivery plan
   e. Mission operations plan
      i. Mission dates
      ii. Total sorties
      iii. Sorties per day
      iv. Time per sortie
      v. Daily schedule
         1. Sunrise/sunset/civil twilight/civil sunrise
         2. Airfield hours
         3. Report times
   f. Training
      i. Initial certification
      ii. Currency
      iii. Checkrides
      iv. Spray Mx
   g. Spray test requirements
   h. DGPS requirements
   i. PPR#
   j. Weather requirements
      i. Weather shop hours
      ii. Winds
      iii. Air stability
      iv. Sky conditions
      v. Humidity
   k. ATC and range clearance
      i. Range schedule
      ii. Conflicts
      iii. NOTAMs
   l. Spray parameters
      i. Groundspeed
      ii. Altitude
      iii. Swath width
      iv. Application rate
      v. flow rate
      vi. psi
   m. Flush and/or purge procedures
2. Mission and personnel coordination items

a. Aircrew, CPMP, spray mx
b. Mission Plans
c. Status of orders
d. Billeting (non-availabilities)
e. Transportation

3. Deployment equipment
   a. DGPS
   b. Trackers
   c. Laptops
   d. Radios and phones
   e. Wx stations
   f. Maps
   g. Swath marking gear
   h. Classified
4. MASS
   a. Configuration
   b. Booms/nozzles
   c. Calibration equipment
5. Deployment installation
   a. In-briefing
   b. Aircraft parking
   c. Security
   d. Loading location
   e. Spill management
   f. HAZMAT disposal
   g. Intelligence
6. Post-mission
   a. De-brief (who/when/where)
   b. Cleanup and cargo assembly
   c. Vehicle and equipment return
   d. AFRC daily SITREP
      i. Acreage
      ii. Volume Chemical
      iii. Total flight time
      iv. Changes to planned schedule
      v. Aircraft Status
   e. Post mission forms return to spray office:
      i. Aerial spray mission report completed
      ii. Copies of 781's
      iii. Loadmaster spray data files

Mission improvement recommendations

INSTALLATION BRIEFING GUIDE

1. Attendees (as required)
   a. Commander/designated representative
   b. CPMP, MC, AC, and NAV (As required)
   c. Project POC
   d. Base operations
   e. Range
   f. Weather
   g. Public Affairs
   h. Fire and medical
   i. JAG
2. Mission objectives
   a. Pest
   b. Pesticide
      i. Diluent
      ii. Application rate
      iii. Toxicity
      iv. Environmental concerns
   c. Operations
      i. Days
      ii. Times
      iii. Number of missions/day
   d. Spray flight parameters
      i. Swath width
      ii. Altitude
      iii. Ground speed
   e. Ground and flight coordination
      i. Communication
      ii. Swath marking
      iii. Ground personnel on range
   f. Chart review
      i. Spray area
      ii. Acreage
      iii. Boundaries
      iv. Sensitive areas
      v. Priority areas
      vi. Validation
         1. Signature
         2. Date
3. Installation Coordination
   a. Weather
      i. Winds
      ii. Temperature
      iii. Humidity
      iv. Ceiling and vis
      v. Air stability
      vi. Precipitation
   b. ATC and range coordination
      i. Restricted areas

ii. Flight hazards
iii. Requested callouts
iv. Schedule conflicts
1. Airspace
2. Airfield operations
v. Routing and flight paths
vi. Priority
vii. NOTAM
c. Load/download
i. Pesticide volume
ii. Diluent volume
iii. Equipment needed
1. Hoses
2. Barrels
3. Mixing stations
4. Download containers
iv. Load team
v. Load area
vi. Spill management plan
vii. HAZMAT disposal
d. Public Affairs
i. Public notification
1. Newspaper
2. TV
3. Radio
4. Base email
ii. Beekeeper notification
iii. Media coordination
e. Legal
i. Hold harmless agreements
ii. Environmental assessments, permits, and waivers
iii. Label directives
f. Medical
i. Emergency procedures
ii. Load team treatment options
iii. Vector monitoring
g. Fire department
i. Pesticide and diluents MSDS
ii. Volatility
iii. PPE requirements
h. Security
i. Flight line access
ii. Aircrew and ground lists
i. Personnel
i. Billeting
ii. Transportation
iii. Messing
j. Intel

Debrief (who, when, where)

PART TWO – NORMAL PROCEDURES

BEFORE SPRAY CHECKLIST

NOTE

This checklist will be initiated by the pilot and will be completed prior to descending for the swath.

1.	**"**Crew, before spray checklist**"**	(PF)
	"Acknowledged**"**	(LM)
2.	Crew briefing - "Complete"	(PF)
3.	Spray area - "Reviewed"	(P,CP,N)
4.	Stall speeds - "Checked"	(P,CP, E)

NOTE

Stall speeds for bank angles of 0, 30, 45, and 60 degrees will be posted.

5.	Altimeters - "Set, state setting"	(P,CP,N)
6.	Radar altimeter - "Set, state setting"	(P,N)
7.	GCAS - "Set, state setting"	(PM)
8.	Communication radios - "Set"	(CP)
9.	DGPS - "Set"	(N)
10.	TD control valve switches - Locked	(E)
11.	Synchrophaser master switch - Off	(E)
12.	Propeller ice control switches - Off	(E)
13.	Seat belt/shoulder harness - "Fastened, unlocked"	(P, CP, E)
14.	NVGs (As Required) – "ON, Ready"	(P,CP,E,N,LM)
15.	Pressurization - As required	(E)
16.	Before Spray checks - "Complete"	(LM, E)

AFTER SPRAY CHECKLIST

1.	Synchrophase master switch - As required	(E)
2.	TD control valve switches - Auto	(E)
3.	Propeller ice control switches - On	(E)
4.	Pressurization - As required	(E)
5.	After spray checks - "Complete"	(LM,E)

PART THREE – EMERGENCY PROCEDURES

CHEMICAL LEAK/SPILL*

**LEAK* is defined as a quantity of chemical which can be contained within the MASS. *SPILL* is defined as a quantity of chemical in excess of the amount which can be contained within the MASS.

1. Notify Pilot – **"SPRAY OFF"**

- Brief description of malfunction (LM)

2. Initiate Immediate Shallow Climb to MSA (Spill only) - (PF)

3. Pilot – **"OXYGEN/ON, 100%"** (if needed) - (ALL)

4. CHEMICAL LEAK/SPILL CHECKLIST – **"COMPLETE"** (ALL)

EMERGENCY CLIMB

1. Announce – **"EMERGENCY CLIMB"** (ANY)

2. Execute – GCAS Escape Maneuver (PF)

- As outlined in TO 1C-130H-1